UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Formulas and symbols

Every formula in scope, then a glossary of every symbol used.

Class 014 · Applications of LDEs: LCR circuits · compiled 17 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S014·8	Unit 2 deck 014 (“Applications of DE – LCR circuit”), slide 8 (rendered .pptx)
T014 00:04:29	Unit 2 AV-summary transcript “Applications of higher-order linear differential equations – RLC Circuits” (11.6 min), segment start
CW U2 P34	Unit 2 class-work problem 34
ESA Jan-24 s3 3b	Past paper: UE23MA141A Jan 2024 set 3, question 3.b

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A Formula sheet

1 Modelling the series LCR circuit

E1.1 · Voltage drops

ESA ★★

$$E_L = L\frac{dI}{dt}, \qquad E_R = RI, \qquad E_C = \frac{Q}{C}$$

GIVES The voltage across the inductor, resistor and capacitor.

APPLIES Any circuit with these elements.

UNITS volts; L in H, R in Ω, C in F, I in A, Q in C

NOTE $E_R = RI$ is Ohm's law.

S014·5 · T014 00:01:18 · T014 00:01:34

E1.2 · Kirchhoff's voltage law

ESA ★★

$$E(t) = V_L + V_R + V_C = L\frac{dI}{dt} + RI + \frac{Q}{C}$$

GIVES The circuit equation in I and Q.

APPLIES Series circuit.

NOTE Sum of the drops = the EMF supplied.

S014·5 · S014·6 · T014 00:01:51 · T014 00:02:21

E1.3 · The LCR equation for the charge

ESA ★★

$$L\frac{d^2Q}{dt^2} + R\frac{dQ}{dt} + \frac{Q}{C} = E(t), \qquad I = \frac{dQ}{dt}$$

GIVES A second-order linear DE with constant coefficients for $Q(t)$.

APPLIES Series LCR circuit; L, R, C constant.

NOTE Divide by L before finding the AE: $Q'' + \frac{R}{L}Q' + \frac{1}{LC}Q = \frac{E}{L}$.

S014·6 · S014·7 · S014·14 · T014 00:02:58 · ESA Jan-24 s3 3b · ESA Jan-24 s4 3b

E1.4 · Initial conditions

ESA ★★

$$Q(0) = Q_0, \qquad I(0) = Q'(0) = I_0$$

GIVES The two conditions that fix c_1, c_2.

APPLIES Usually $Q_0 = I_0 = 0$ (switch closed at $t = 0$).

NOTE Use $Q(0)$ in $Q(t)$, then differentiate $Q(t)$ and use $I(0)$.

S014·7 · S014·11 · S014·12 · T014 00:03:41 · T014 00:08:21

E1.5 · The equation for the current

ESA ★

$$L\frac{d^2I}{dt^2}+R\frac{dI}{dt}+\frac{I}{C}=\frac{dE}{dt},\qquad LI'(0)=E(0)-RI(0)-\frac{Q(0)}{C}$$

GIVES $I(t)$ directly, by differentiating E1.3 once.

APPLIES The current is asked and E is differentiable.

NOTE The second condition comes from E1.2 at $t=0$. A constant E gives $\frac{dE}{dt}=0$.

ADDED FOR CLARITY S014·5 · S014·16

2 Solving the circuit equation: CF, PI and damping

E2.1 · Auxiliary equation and its roots

ESA ★★

$$Lm^2+Rm+\frac{1}{C}=0,\qquad m=\frac{-R\pm\sqrt{R^2-4L/C}}{2L}$$

GIVES The CF (transient) of the charge.

APPLIES Every LCR problem.

NOTE Slide example: $2m^2+8m+10=0$, $m=-2\pm i$.

S014·9 · S014·15 · T014 00:06:48

E2.2 · Damping regimes

ESA ★

$$R^2>\tfrac{4L}{C}:\ Q_{tr}=c_1e^{m_1t}+c_2e^{m_2t}$$
$$R^2=\tfrac{4L}{C}:\ Q_{tr}=(c_1+c_2t)e^{-Rt/(2L)}$$
$$R^2<\tfrac{4L}{C}:\ Q_{tr}=e^{-Rt/(2L)}(c_1\cos\omega_dt+c_2\sin\omega_dt)$$

GIVES The form of the transient.

APPLIES Compare R^2 with $\frac{4L}{C}$.

NOTE Over-, critically, underdamped. All decay: $m_1,m_2<0$ or real part $-\frac{R}{2L}<0$.

ADDED FOR CLARITY S014·9 · S014·15

E2.3 · Damped frequency

ESA ★

$$\omega_d=\sqrt{\frac{1}{LC}-\frac{R^2}{4L^2}},\qquad \omega_0=\frac{1}{\sqrt{LC}}\ (R=0)$$

GIVES The angular frequency of the underdamped transient.

APPLIES $R^2<\frac{4L}{C}$.

UNITS rad/s

NOTE ESA s3: $\sqrt{625-400}=15$; s4: $\sqrt{500-100}=20$.

ADDED FOR CLARITY S014·9 · ESA Jan-24 s3 3b

E2.4 · Solution = transient + steady state ESA ★★

$$Q(t) = \underbrace{\text{CF}}_{\text{transient}} + \underbrace{\text{PI}}_{\text{steady state}}, \qquad E = \text{const} \Rightarrow \text{PI} = CE$$

GIVES The structure of every answer.

APPLIES Any $E(t)$.

NOTE Constant E: $\frac{E/L}{D^2+\frac{R}{L}D+\frac{1}{LC}}$ with $D \to 0$ gives CE (slide: $\frac{5}{5} = 1 = 0.1 \times 10$).

S014·10 · T014 00:07:37 · T014 00:07:52

E2.5 · Steady-state amplitude (impedance) ESA ★

$$E = E_0 \sin \omega t: \quad I_{ss} = \frac{E_0}{Z}\sin(\omega t - \theta), \quad Z = \sqrt{R^2 + \left(L\omega - \frac{1}{C\omega}\right)^2}, \quad |Q_{ss}| = \frac{E_0}{\omega Z}$$

GIVES A check on any PI.

APPLIES Sinusoidal source.

NOTE ESA s3: $Z = \sqrt{1600 + 120^2} = 126.5$, $|Q_{ss}| = \frac{100}{5 \cdot 126.5} = 0.158 = \sqrt{0.15^2 + 0.05^2}$.

ADDED FOR CLARITY ESA Jan-24 s3 3b · S014·15

3 Past-paper circuits: AC sources and free discharge

E3.1 · PI of a sinusoidal source (Type 2) ESA ★★

$$\frac{1}{D^2 + \frac{R}{L}D + \frac{1}{LC}}\cos \omega t = \frac{\cos \omega t}{\left(\frac{1}{LC} - \omega^2\right) + \frac{R}{L}D} \quad \text{then rationalise}$$

GIVES The steady-state charge.

APPLIES $E = E_0 \sin \omega t$ or $E_0 \cos \omega t$; replace D^2 by $-\omega^2$.

NOTE No failure while $R > 0$ (the CF decays, the forcing does not).

ESA Jan-24 s3 3b · ESA Jan-24 s4 3b · S014·15

4 First-order circuits: RL and RC

E4.1 · RL circuit with a sinusoidal source ESA ★★

$$L\frac{di}{dt} + Ri = E\sin\omega t,\ i(0) = 0:$$
$$i = \frac{E}{\sqrt{R^2 + L^2\omega^2}}\left[\sin(\omega t - \varphi) + e^{-Rt/L}\sin\varphi\right],\ \varphi = \tan^{-1}\frac{L\omega}{R}$$

GIVES The current in an RL circuit.

APPLIES No capacitor; switch closed at $t = 0$.

NOTE Transient $e^{-Rt/L}$, time constant $\frac{L}{R}$.

CW U2 P34

E4.2 · RC circuit with a cosine source ESA ★★

$$R\frac{di}{dt} + \frac{i}{C} = \frac{dE}{dt},\ E = E_0\cos pt,\ i(0) = 0:$$
$$i = \frac{CpE_0}{1 + p^2R^2C^2}\left[RCp\cos pt - \sin pt - pRC\,e^{-t/(RC)}\right]$$

GIVES The current in an RC circuit.

APPLIES No inductor.

NOTE From $Ri + \frac{Q}{C} = E$ differentiated; time constant RC.

CW U2 P35

B Symbol glossary

Symbol	Name	Meaning	Unit	Appears in
L	Inductance	Henries (H)	H	E1.1, E1.3
R	Resistance	Ohms (Ω)	Ω	E1.1, E1.3
C	Capacitance	Farads (F); not an integration constant	F	E1.1, E1.3
$E(t)$	Electromotive force	Source voltage; E, E_0 also amplitudes (CW P34–35)	V	E1.2, E4.1
Q, q	Charge	On the capacitor; slides 13–16 write q	C (coulomb)	E1.3
I, i	Current	$\frac{dQ}{dt}$	A	E1.2, E1.5
V_L, V_R, V_C	Voltage drops	Also E_L, E_R, E_C on slide 014·5	V	E1.1, E1.2
$\dot{q}$, $\ddot{q}$	Time derivatives	$\frac{dq}{dt}$, $\frac{d^2q}{dt^2}$ (slides 13–16)		E1.3
c_1, c_2	Arbitrary constants	Of Q; slide 014·16's C_1, C_2 are constants of i		E2.2
ω_d	Damped frequency	$\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$	rad/s	E2.3
ω, p	Source frequency	In $\sin \omega t$, $\cos pt$	rad/s	E2.5, E4.1, E4.2
Z	Impedance	$\sqrt{R^2 + (L\omega - \frac{1}{C\omega})^2}$	Ω	E2.5
φ	Phase angle	$\tan^{-1} \frac{L\omega}{R}$ (RL circuit)		E4.1

Remark. The deck mixes Q, I (slides 4–12) with q, i, $\dot{q}$ (slides 13–16); they mean the same. E is both the source voltage $E(t)$ and, in CW P34–P35, the amplitude E or E_0 of a sinusoid. C (capacitance, farads) is never an integration constant here: those are c_1, c_2 (or C_1, C_2 on slide 014·16, which are constants of the current). ω and p are forcing frequencies; φ is a phase angle.